complicated. Think of it this way: the space of all possible wave functions, Hilbert space, is very big. It's not any bigger in Many-Worlds than in other versions of quantum theory; it's precisely the same size, and that size is more than big enough to describe a large number of parallel realities. Once you can describe superpositions of spinning electrons, you can just as easily describe superpositions of universes. If you're doing quantum mechanics at all, the potential for many worlds is there, and ordinary Schrödinger evolution tends to bring them about, like it or not. Other approaches just choose to somehow not make use of the full richness of Hilbert space. They don't want to accept the existence of other worlds, so they need to work hard to get rid of them somehow."

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"Fine," muttered her father, not fully convinced but apparently ready to move on to the next question. He took a sip of his drink and peered at his phone. "Isn't there also a philosophical problem with the theory? I'm no philosopher myself, but Karl Popper and I both know that a good scientific theory is supposed to be falsifiable. If you can't even imagine an experiment that might prove your theory wrong, it's not really science. That's exactly the situation with all these other worlds, isn't it?"

"Well, yes and no."

"That's the go-to answer to any philosophy question."

"The price we pay for being notorious sticklers for precision." Alice laughed. "Sure, Popper had this proposal that scientific theories must be falsifiable. It was an important idea. But in the back of his mind he was thinking about the difference between theories such as Einstein's general relativity, which made definite empirical predictions for the bending of light by the sun, and those like Marxist history or Freudian psychoanalysis. The problem with the latter ideas, he thought, was that no matter what actually happened, you could cook up a story to explain why it was so."

"That's what I thought. I haven't read Popper myself, but I appreciate that he put his finger on something crucial about science."

Alice nodded. "He did. But to be honest, most modern philosophers of science agree that it isn't the complete answer. Science is messier than that, and what separates science from non-science is a subtle issue."